

Inductive Bias:

- ↳ Inductive Bias (or the inherent bias of the algorithm) are assumptions that are made by the learning algorithm to form a hypothesis or generalization ~~but~~ beyond the set of training instances in order to classify unseen data.
- ↳ Inductive Biases play an important role in the ability of machine learning models to generalize the unseen data.
- ↳ A strong inductive bias can lead our model to converge to the global optimum.
- ↳ On the other hand, a weak inductive bias can cause the model to find only local optima and can be greatly affected by the random changes in the initial states.
- ↳ Thus in machine learning, inductive bias refers to the assumption ~~error~~ or preconceptions that a model or algorithm makes about the underlying distribution of data.
- ↳ These biases can influence the model's ability

to learn from a given dataset and can affect the performance of the model on new, unseen data.

↳ we can categorize inductive biases into two different groups called relational and non-relational. The former represents the relationship between entities in the network, while the latter is a set of techniques that further constrain the learning algorithm.

↳ There are several ways to describe the inductive bias of a model, including:

↳ The choice of model architecture

↳ Selection of features

↳ The type of regularization applied to the model

↳ It is important to consider the inductive bias of a model when selecting or designing a machine learning algorithm, because it can have a significant impact on model's performance.

Ex: linear regression model has an inductive bias towards linear relationship b/w variables while, a decision tree has a inductive bias

Inductive Bias

↳ In machine learning, the term inductive bias refers to a set of (explicit or implicit) assumption made by a learning algorithm in order to perform induction,

is that is, to generalize a finite set of observation (training data) into a general model of the ~~max~~ domain.

↳

- Preference : Impose ordering on hypothesis space

Inductive learning

Inducing a general function from training examples.

- Construct hypothesis h to agree with c on the training example.
- A hypothesis is consistent if it agrees with all training examples.
- A hypothesis said to generalize well if it correctly predicts the value of y for novel example.

Inductive Learning is an /// Posed Problem :

Unless we see all possible examples the data is not sufficient for an inductive learning algorithm to find a unique solution.

Inductive Learning Hypothesis

Any hypothesis h found to approximate the target function c well over a sufficiently large set of training examples D will also approximate the target function well over other unobserved examples.

Learning as Refining the Hypothesis Space

Concept learning is a task of searching an hypotheses space of possible representations looking for the representation(s) that best fits the data, given the bias.

The tendency to prefer one hypothesis over another is called bias.

Given a representation, data, and a bias, the problem of learning can be reduced to one of search.